CS & IT ENGINEERING

Operating System

Memory Management

Lecture -2

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Recap of Previous Lecture

doubt ?

Topic Memory Management

Topic Contiguous Memory Management Technique

Topics to be Covered

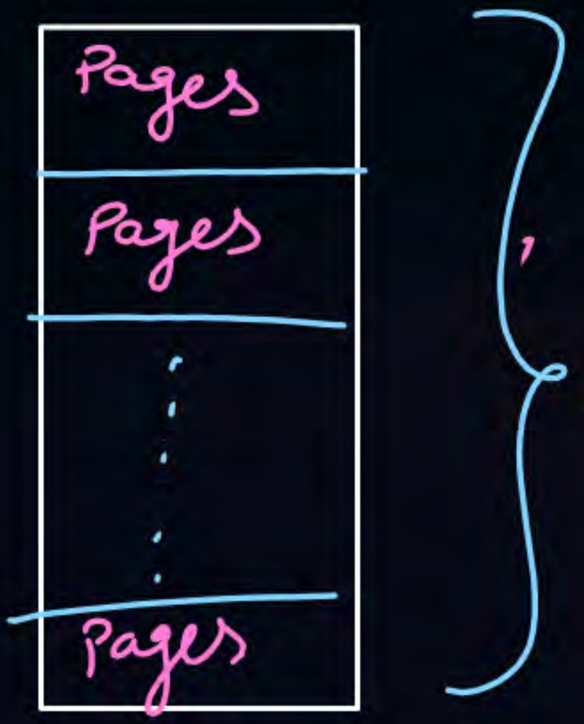
Topic Non-Contiguous Memory Management Technique

- Paging
- segmentation

Topic Paging

Topic Address Translation

Topic : Paging



equal size partitions of process

Topic : Paging

- Process is divided in equal size of partitions called as pages
- Physical memory is divided in same size of partitions called as frames
  main
- Pages are scattered in frames

Topic : Paging

Example:

P1
Process

Page 0
Page 1
Page 2
Page 3

4 pages

Page table (mapping of which page stored on which frame)

0	2
1	6
2	4
3	1

frame no.

4 entries (one for each page)

P2

0	5

Main memory

	frame
	0
Page 3	1
Page 0	2
	3
Page 2	4
Page 0 for Process P2	5
Page 1	6
	7

8 frames

Topic : Paging

Consider

- A process has 4 pages
- Main memory has 8 frames

Process

00	
01	
10	
11	

Page table

00	010
01	110
10	100
11	001

m.m.

000	
001	Page 11
010	Page 00
011	
100	Page 10
101	
110	Page 01
111	

Topic : Paging

Page Size = 2 Bytes

logical address

Process

logical address	Process	Page
000	a	00
001	b	00
010	c	01
011	d	01
100	e	10
101	f	10
110	g	11
111	h	11

Page Table

Page	Frame
00	010
01	110
10	100
11	001

mm address (Physical address)

Physical Memory

Address	Content	Frame
0000		frame 000
0001		
0010	g	001
0011	h	
0100	a	010
0101	b	
0110		011
0111		
1000	e	100
1001	f	
1010		101
1011		
1100	c	110
1101	d	
1110		111
1111		

Process = 4 Pages = $4 * 2B = 8$ Bytes $= 2^3 B$

mm = 8 frames = $8 * 2B = 16$ Bytes $= 2^4 B$

CPU generates byte no. $\Rightarrow (110)_2$

11	0

page no $= (11)_2$

byte no $= (0)_2$

search into P.T. = frame no $\Rightarrow 001 \Rightarrow$ access byte 0 of frame 001

001	0

mm address

CPU provides byte no.

→ Page no. → Search page table → frame no.

→ which byte of this page → which byte

Page number	byte no. of the page
p	d

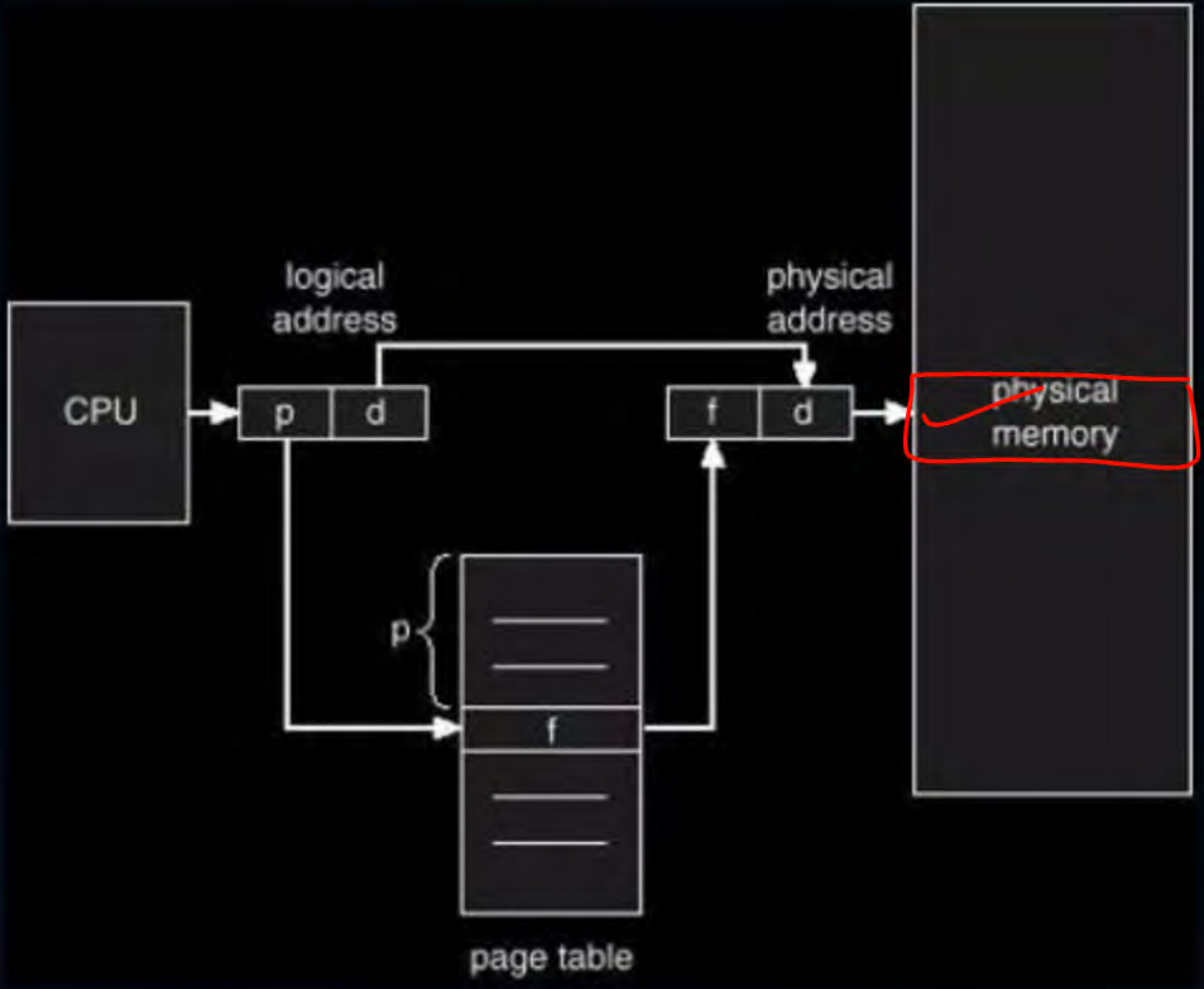
mm address

frame no.	byte no. of the page
f	d

→ Logical address space ⇒ process size

→ physical —||— ⇒ physical mem. size

Topic : Paging



Topic : Paging

- Processor will have a view of process and its pages
- Page table is used to map a process page to a physical frame
- Number of entries in page table = Number of pages in process
- OS maintains a page table for each process
- page table size = no. of entries in P.T. * 1 entry size
  = no. of pages in process * 1 entry size
- 1 page table entry size = frame no. bits + extra bits

if no of pages in process = 2^P, then no. of bits for page number = P bits

— 〃 — frames in mm = 2^f, — 〃 — frame number = f bits

no. of pages in process = $\dfrac{\text{Process size}}{\text{Page size}}$

no. of frames in mm = $\dfrac{\text{mm size}}{\text{Page size}}$

no. of bits for byte no. (offset)

$= \log_2(\text{Page size})$

examples:-

L.A. = 15 bits

P.A. = 20 bits

Page size = 128 bytes $= 2^7 B \Rightarrow d = 7$ bits

L.A. ← 15 →

p	d
8	7

P.A. ← 20 →

f	d
13	7

no. of pages in process $= 2^8 = 256$

no. of frames in m.m. $= 2^{13} = 8K$

no. of entries in P.T. $= 256$

extra bits

P.T. Size $= 256 * (13 + 0)$ bits

Topic : Question

#Q. Consider a paged memory system where the logical address of 29 bits and physical address of 35 bits. If each page table entry is of 4 bytes and page size is 1KB then:

32 bits

1KB $\rightarrow 2^{10}B \Rightarrow d = 10$ bits

1. Number of pages in process? 2^{19}
2. Number of frames in main memory? 2^{25}
3. Number of bits for page number? 19
4. Number of bits for frame number? 25
5. Number of entries in page table? 2^{19}
6. Page table size? $2^{19} * 4B = 2^{21}B = 2MB$

LA

29

p	d
19	10

PA

35

f	d
25	10

Ques) In prev questⁿ no. of bits (extra bits) used in page table for each page table entry for protection ?

Solⁿ

$$32 \text{ bits} = 25 + \text{extra}$$

$$\text{extra} = 7 \text{ bits}$$

Topic : Question

[GATE-2001]

2^{26} B ⇒ P.A. = 26 bits

#Q. Consider a machine with 64 MB physical memory and a 32-bit virtual (logical) address space. If the page size is 4 KB, what is the approximate size of the page table?

A) 16 MB

B) 8 MB

C) 2 MB ✓

D) 24 MB

LA

32	
p	d
20	12

PA.

26	
f	d
14	12

Page size = 4KB
$= 2^{12}$ B
⇓
d = 12 bits

P.T. size $= 2^{20} * (14 + 0)$ bits
≅ 2MB

$2^{24}B = L.A = 24$ bits

#Q. Consider a paged memory system where the process size is 16MB and main memory size is 4GB. The page size is 2KB. $\longrightarrow 2^{11}B \Rightarrow d = 11$ bits

$\hookrightarrow 2^{32}B \Rightarrow P.A. = 32$ bits

1. Number of pages in process? 2^{13}
2. Number of frames in main memory? 2^{21}
3. Number of bits for page number? 13
4. Number of bits for frame number? 21
5. Number of entries in page table? 2^{13}
6. Page table size? $2^{13} * 21$ bits

LA 24

p	d
13	11

PA 32

f	d
21	11

H.W

#Q. Consider a paged memory system where the process size is 128MB and main memory size is 2GB. The page size is 1KB.

1. Number of pages in process?
2. Number of frames in main memory?
3. Number of bits for page number?
4. Number of bits for frame number?
5. Number of entries in page table?
6. Page table size?

H.W

#Q. Consider a paged memory system where the logical address is 25 bits and physical address is 33 bits. The page size is 4KB.

1. Number of pages in process?
2. Number of frames in main memory?
3. Number of bits for page number?
4. Number of bits for frame number?
5. Number of entries in page table?
6. Page table size?

[MCQ] [GATE-2015]

frame no. $d = 13$ bits

#Q. A computer system implements 8 kilobyte pages and a 32–bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the logical address supported by the system is <u>36</u> bits?

L.A.

p	d
23	13

$\underbrace{23 + 13}_{36 \text{ bits}}$

P.A. $\longleftarrow 32 \longrightarrow$

f	d
19	13

P.T. size = no. of pages * 1 P.T. entry size

$24MB = \text{no. of pages} * (1+1+3+19)$ bits

$\cancel{24}M * 8 \cancel{\text{bits}} = \text{no. of pages} * \cancel{24} \cancel{\text{bits}}$

$\text{no. of pages} = 8M = 2^{23} \Rightarrow p = 23$ bits

2 mins Summary

Topic Non-Contiguous Memory Management Technique

Topic Paging

Topic Address Translation

Happy Learning

THANK - YOU